Elderly Care Management System

Comprehensive High-Fidelity UI/UX Prototype, Design System Tokens, and Interface Architecture for Geriatric Home Healthcare

ACADEMIC SUBMISSION PROFILE

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Project Title: Elderly Care Management System
Design Tooling: Figma / Canva Vector Grade Architecture
Date: October 2026

DESIGN SYSTEM OBJECTIVES

- **Senior-Centric Accessibility:** WCAG AAA (≥7:1) contrast, minimum **14pt (18.67px)** body text on Senior UI (14px minimum across clinical portals), & ≥56px touch targets.
- **5 Role-Isolated Portals:** Dedicated interfaces for Family Members, Caregivers, Senior Residents, Physicians, and Administrators.
- **Zero-Error Safety & SOS:** Color-coded eMAR adherence, shift handoff logs, and 1-tap 999 SOS dispatch.

SEQUENCED INTERFACE SPECIFICATION (15 PAGES • 12 DEDICATED SCREENS)

1. Design System Tokens & Accessibility
2. Multi-Role Screen Flow & IA
3. Screen 1: Multi-Role Authentication
4. Screen 2: Central Family Dashboard
5. Screen 3: Elderly Profile & Health Record
6. Screen 4: Medicine Schedule & Tracker
7. Screen 5: Add Medication Form Modal
8. Screen 6: Doctor Appointment Tracking
9. Screen 7: Daily Care Routine Checklist
10. Screen 8: Caregiver Shift Handoff Log
11. Screen 9: Instant Emergency SOS Portal
12. Screen 10: Senior Resident Dashboard
13. Screen 11: Physician EHR & Consult Portal
14. Screen 12: Admin Governance & Audit Logs

01. Design System & Design Tokens

Color Palette, Typography Scale, UI Components, and Accessibility Standards

Design System v2.0 | Figma Tokens

Core Medical Color Tokens

Teal Primary
#0D9488 • Brand & CTAs

Deep Slate
#0F172A • High Contrast Text

Amber Alert
#F59E0B • Med Reminders

Emergency Red
#E11D48 • SOS & Paramedics

Reconciled Typography Hierarchy

Heading 1 (24px / 18pt) — Portal & Section Titles

Heading 2 (20px / 15pt) — Card Headers & Modals

Senior Body (14pt / 18.67px min) — Accessible Senior UI

Clinical Body (14px / 10.5pt min) — EHR & Staff Tables

Interactive Badges & Buttons

✓ Dose Completed | ⏳ Due in 25m | 🚨 Critical SOS | 🩺 Physician Review

Primary Action | **Secondary Option**

Emergency Dial

Reconciled Geriatric & Elderly-Friendly UX Guidelines (WCAG 2.1 AAA)

- **Minimum Body Size:** Strict **14pt (18.67px)** minimum body text in Senior Resident UI; **14px** minimum on dense clinical tables.
- **Touch Targets:** Minimum **48px** clinical controls; **56px–72px+** buttons in Senior Resident UI.
- **Dual Encoding & Contrast:** ≥7:1 contrast ratio; every state pairs color + icon + text label.

TYPOGRAPHY RECONCILIATION

Reconciles senior accessibility and clinical density under one rulebook: **14pt (18.67px)** minimum body floor across Senior Portal and **14px** minimum across clinical staff interfaces (eliminating legacy 13px copy).

DESIGN TOKENS

#0D9488 Brand | #0F172A Base | 14pt / 18.67px Senior | 14px Clinical Body | ≥56px Senior Touch | Inter / Plus Jakarta

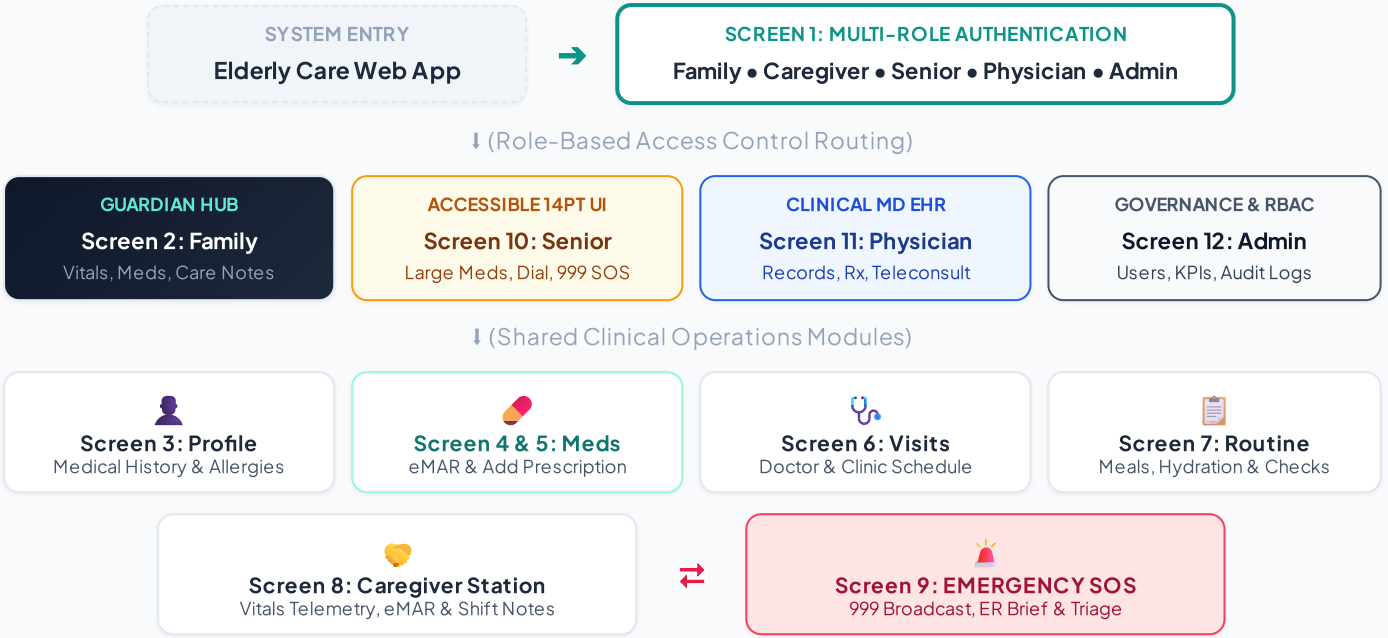
KEY STANDARDS

- Card-based modular layouts across 5 role portals.
- Real-time timestamp & audit trail verification.
- Optimized for tablets, desktops & nurse stations.

02. Screen Flow & Information Architecture

End-to-End User Journey from Authentication to Daily Operations and Emergency Protocol

Architecture | User Journey



ARCHITECTURE STRATEGY

Role-Routed Topology: Each persona lands on an interface tailored to their workflow (accessible large cards for Seniors, EHR workspaces for Physicians, governance/audit tools for Admins).

SCREEN SEQUENCE

- **Screen 1:** 5-role auth gateway
- **Screens 2–7:** Family & daily care ops
- **Screens 8–9:** Caregiver handoff & SOS
- **Screen 10:** Dedicated Senior UI (14pt)
- **Screen 11:** Physician EHR & teleconsult
- **Screen 12:** Admin RBAC & audit logs

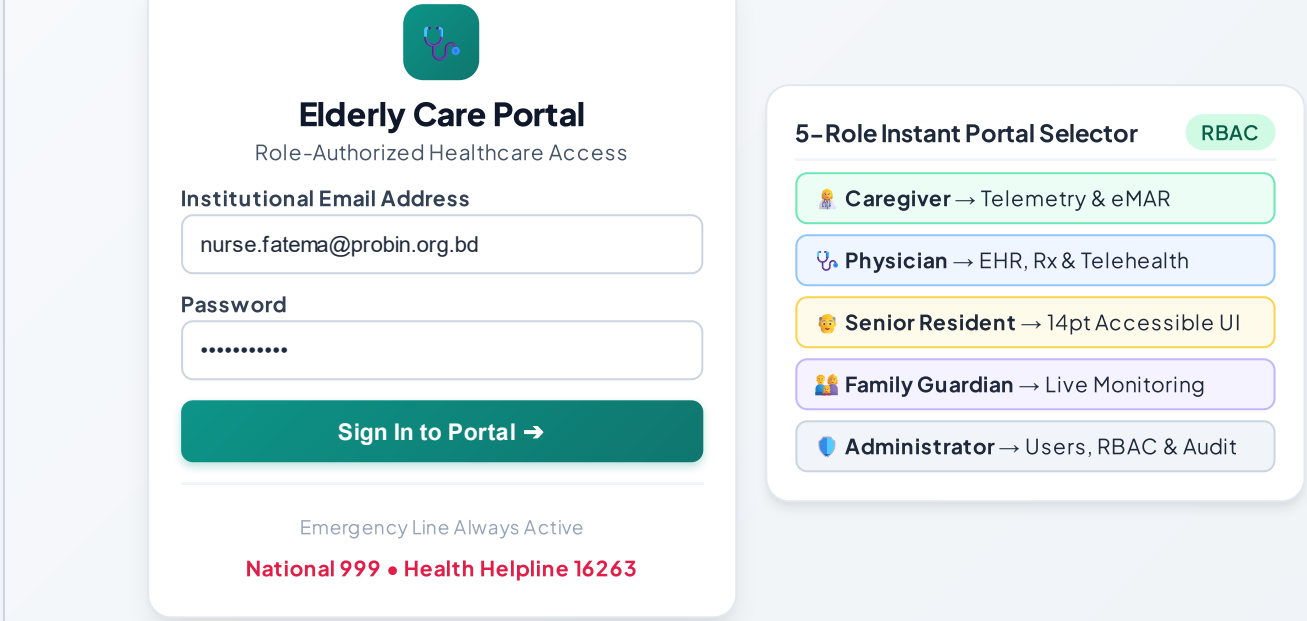
03. Screen 1: Multi-Role Authentication Portal

Secure RBAC login with instant role switching across Family, Caregiver, Senior, Physician & Admin portals

Screen 1 of 12 | Auth & 5 Roles

https://elderlycare.portal/login

SSL 256-bit • JWT RBAC



SCREEN PURPOSE

Routes all 5 clinical and resident personas into strictly isolated interfaces verified via JWT RBAC tokens.

UX DETAILS

- **5-Role Switcher:** Supports Guardian, Caregiver, Senior, Physician & Admin portals.
- **Accessible Entry:** Minimum 48px touch targets & clear contrast.
- **Emergency Access:** 999 / 16263 helplines visible prior to login.

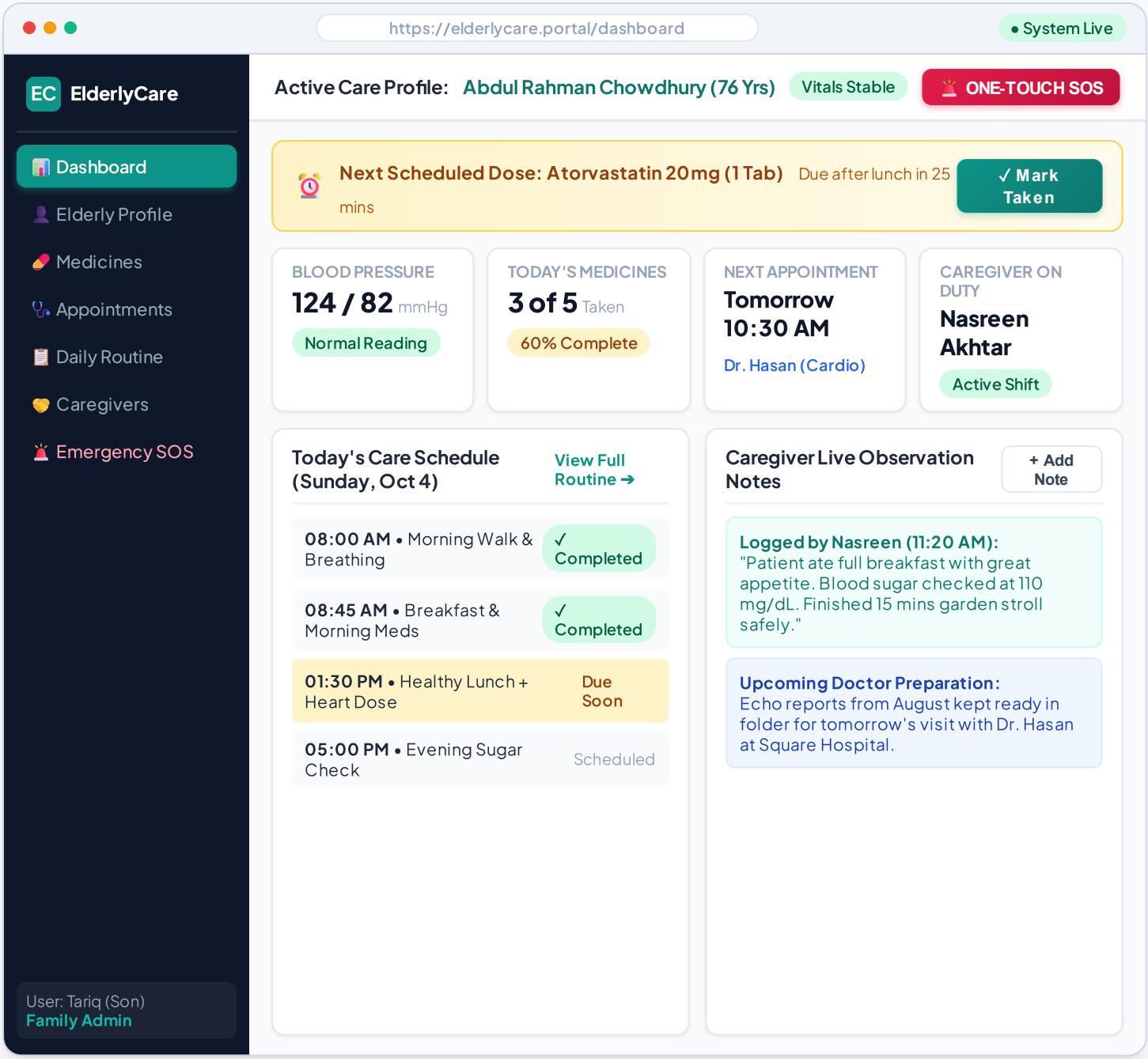
FIGMA TOKENS

#0D9488 Primary | 12px Card Radius | Shadow-LG

04. Screen 2: Central Family Member Dashboard

Comprehensive status command center: vitals summary, upcoming medicine alert, daily routine & SOS

Screen 2 of 12 | Core Command Center



SCREEN OBJECTIVE

Provide family members complete peace of mind in under 5 seconds. Immediate overview of vitals, adherence percentage, and live caregiver shift notes.

UX COMPONENTS

- **Amber Alert Banner:** Highlights the immediate next medication.
- **Stat Cards:** Real-time BP, pill count, and duty nurse status.
- **Instant SOS:** Persistent Red trigger for immediate help.

DATA HANDLED

Vitals Stream | Adherence % | Shift Status

05. Screen 3: Elderly Profile & Medical History

Comprehensive electronic health record: personal demographics, chronic conditions, and physician contacts

Screen 3 of 12 | EHR & Demographics

https://elderlycare.portal/profile/senior-01

Edit Records

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Record ID: EC-2026-881

Elderly Health Profile & Emergency Summary

Penicillin Allergy



Abdul Rahman Chowdhury

Blood Group: B+ (Positive)

Age: **76 Years** • Male • Born: 14 Aug 1950 • Weight: 68 kg • Height: 5' 7"

Residential Address: House 42, Road 11, Dhanmondi R/A, Dhaka-1209

Mobility: Walking Cane

Insurance: Active (Pragati Life)

Chronic Conditions

- Type-2 Diabetes — Since 2012
- Hypertension (High BP) — Since 2015
- Coronary Stent — Placed 2021
- Mild Osteoarthritis — Left Knee

Primary Doctors

Dr. K. M. Hasan
Cardiologist, Square Hospital
Call

Dr. Nusrat Jahan
Endocrinologist, Labaid
Call

Care Directives & Diet

- **Diet:** Strict low-sodium, diabetic-friendly meals (under 1500 kcal).
- **Hydration:** 200ml fluid intake prompt every 2 hours.
- **Fall Prevention:** Non-slip bathroom mats & walking cane assistance mandatory.

SCREEN PURPOSE

A single-pane medical repository for attending clinicians and rotating caregivers, ensuring critical allergies and pre-existing conditions are never overlooked.

UI FEATURES

- **Critical Allergy Tag:** Penicillin alert highlighted in vivid red.
- **Direct Calling:** Fast dial triggers for primary treating physicians.
- **Mobility Flag:** Prepares caregivers for physical support needs.

DATA ARCHITECTURE

Demographics | Allergy Protocol | Physician Links

06. Screen 4: Medicine Schedule & Tracker

Chronological prescription tracker organized by Morning, Afternoon, Evening, and Bedtime slots

Screen 4 of 12 | Medication Safety

https://elderlycare.portal/medications

+ Add Medication

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Daily Dose Target: 5 Pills

Today's Schedule (Oct 4) | All Prescriptions (8) | Refill Alerts (2) | 3 of 5 Taken

MORNING ROUTINE (08:00 AM – 09:00 AM)

✓ All Taken (Verified by Nurse Nasreen)

Metformin 500mg (1 Tablet) — Taken 08:35 AM
With Breakfast • Dr. Nusrat

Amlodipine 5mg (1 Tablet) — Taken 08:35 AM
After Breakfast • Dr. Hasan

AFTERNOON ROUTINE (01:30 PM – 02:00 PM)

⏳ Due in 25 Minutes

Atorvastatin Lipitor 20mg (1 Tablet - Cholesterol)
Take immediately following lunch with water • Prescribed by Dr. Hasan

✓ Mark Taken | Snooze 15m

NIGHT & BEDTIME ROUTINE (09:30 PM – 10:00 PM)

2 Doses Scheduled

Metformin 500mg (1 Tablet) — Upcoming
After Dinner

Calcium D3 (1 Tablet) — Upcoming
With Warm Milk

SCREEN PURPOSE

Eliminates missed or doubled medication doses. Real-time logging creates a verified digital audit trail with nurse verification.

SAFETY CONTROLS

- **Chronological Sorting:** Clear time buckets prevent confusion.
- **One-Click Confirmation:** Fast "Mark Taken" button updates family dashboard instantly.
- **Refill Tracking:** Automatic counter notifies when 5 pills remain.

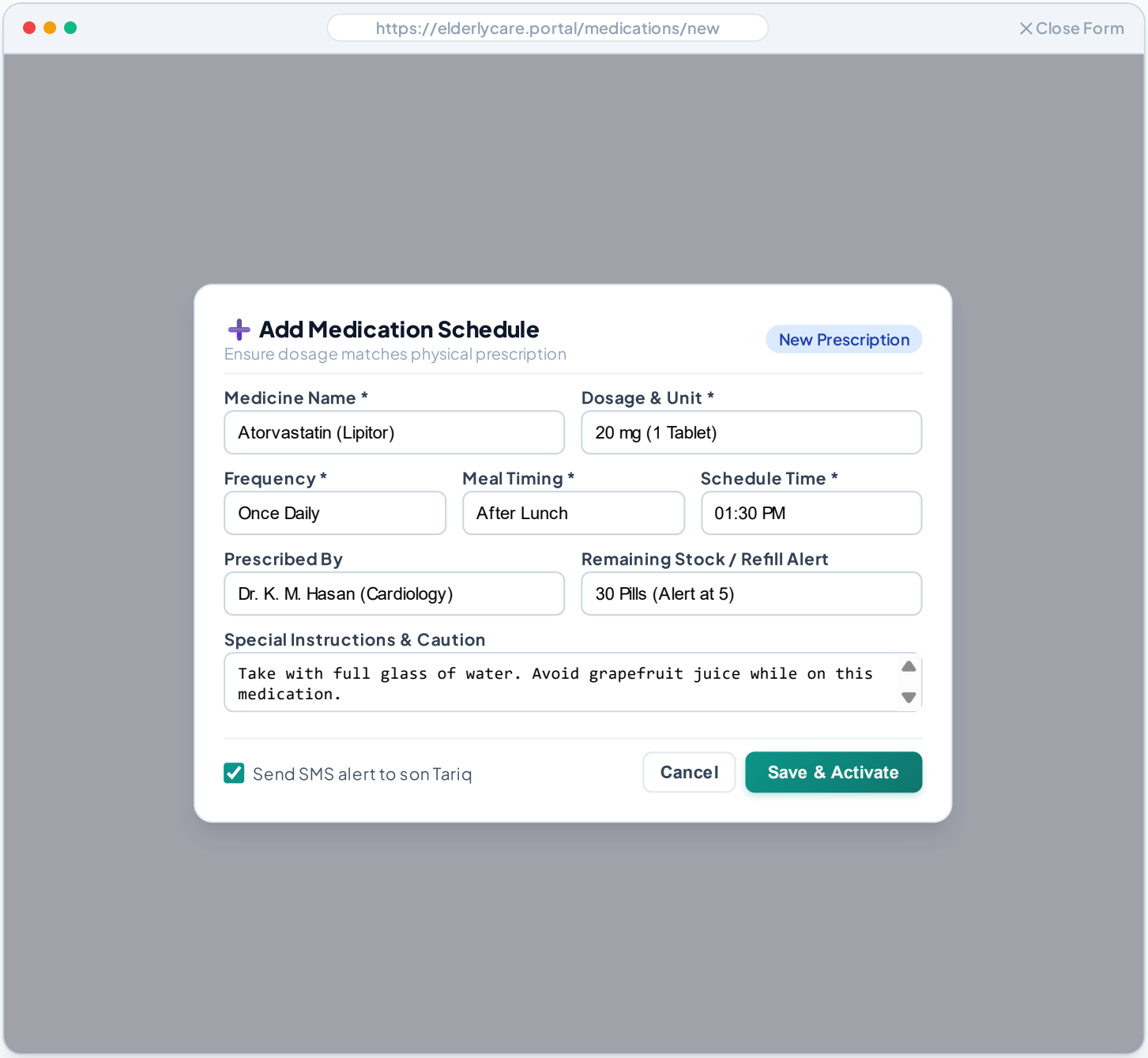
AUDIT TRAIL

Timestamped | Nurse ID | Missed Alert

07. Screen 5: Add / Edit Medication Form

Prescription entry form with dose validation, meal-timing association, and low-stock refill triggers

Screen 5 of 12 | Prescription Input



FORM VALIDATION

Medical safety constraints prevent accidental overdose entries or uncalibrated frequencies.

KEY FIELDS

- **Prescribing Doctor:** Links the dose directly to clinical consultations.
- **Meal Timing:** Before meal / with food / after meal selectors.
- **Inventory Counter:** Automatically computes days until refill needed.

MODAL BEHAVIOR

Backdrop Blur | Required Validation | SMS Hook

08. Screen 6: Doctor Appointment Tracking

Medical appointment calendar, physician directory, consultation reports, and family escort details

Screen 6 of 12 | Clinical Scheduling

https://elderlycare.portal/appointments

+ Book Visit

EC ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Next: Tomorrow 10:30 AM

Doctor Consultation Schedule & History

SMS Alert: 24h & 2h Prior

NEXT SCHEDULED CONSULTATION

Dr. K. M. Hasan — Senior Cardiologist

Square Hospital, Panthapath • **Tomorrow (Monday, Oct 5) at 10:30 AM** (Room 314)

Clinical Goal: 3-Month Echo & ECG Review • Escort: Tariq (Son) • Transport Arranged

View Hospital Map

Notification Active

Future Scheduled Visits

Dr. Nusrat Jahan (Diabetes)

Labaid Diagnostic, Dhanmondi

Oct 18, 2026 • 05:00 PM

(Fasting Blood Sugar)

Reschedule

Dr. Shahed Ali (Orthopedic)

Popular Hospital, Shantinagar

Nov 02, 2026 • 11:00 AM (Knee Joint Mobility)

Reschedule

Past Consultation Records & RX

Routine Cardiology Follow-up

Sep 04, 2026 • Dr. K. M. Hasan

(BP: 120/80)

View RX

Ophthalmology Vision Assessment

Aug 12, 2026 • Vision stable, no cataract surgery

View RX

SCREEN PURPOSE

Centralizes all doctor appointments so family members can coordinate transport, tests, and family escorts well in advance.

FEATURE HIGHLIGHTS

- **Escort Designation:** Clearly designates which son/daughter will attend.
- **Digital Prescriptions:** Instant access to scanned prescriptions from past visits.
- **Map Integration:** Direct clinic navigation directions.

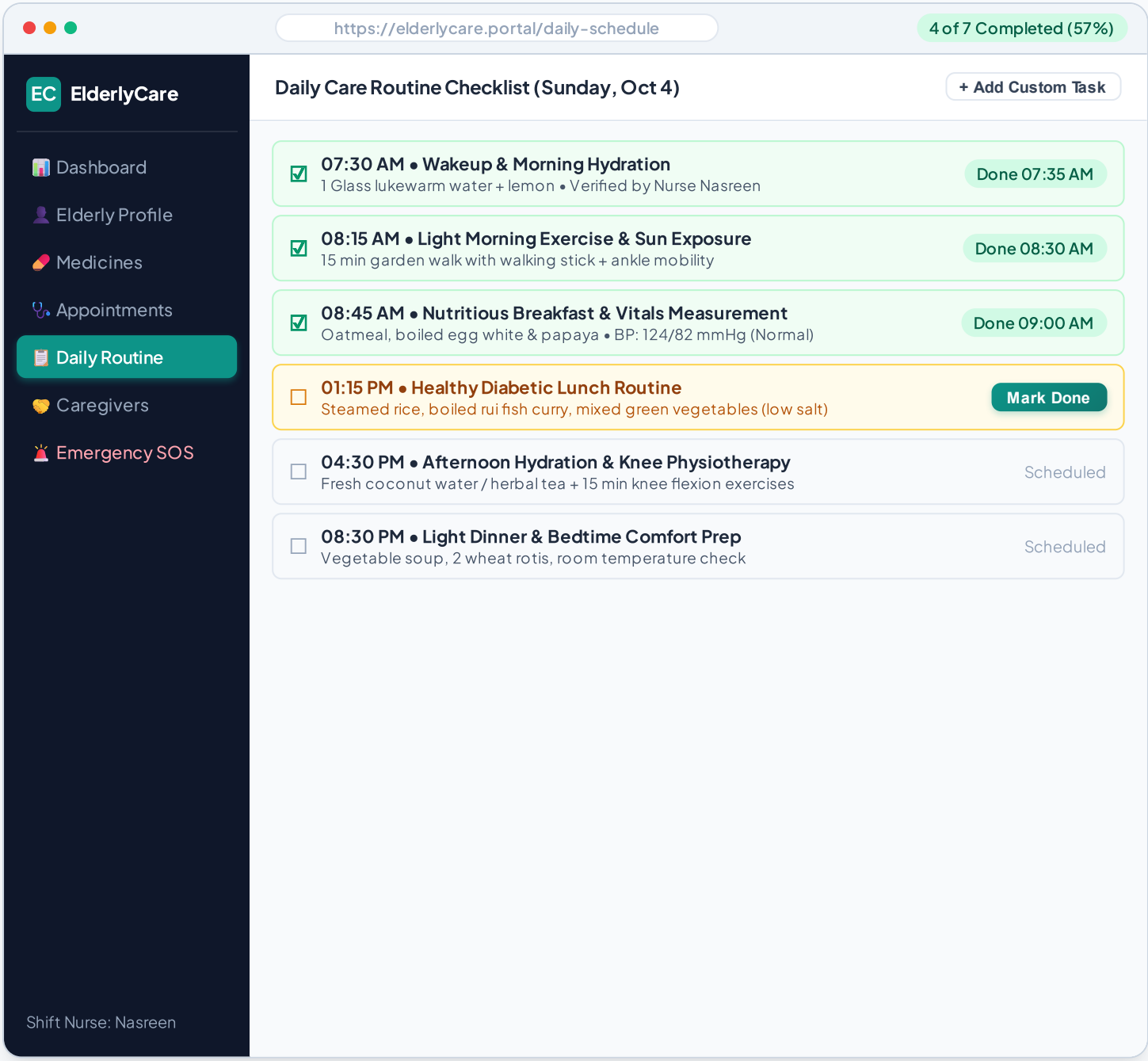
SYSTEM INTEGRATIONS

Google Calendar | SMS Reminders | PDF RX Archive

09. Screen 7: Daily Care Schedule & Activities

Holistic elderly lifestyle checklist: meals, hydration checks, light mobility, and vitals recording

Screen 7 of 12 | Daily Routine Checklist



HOLISTIC HEALTH FOCUS

Comprehensive care requires nutrition, hydration, and mobility, not just pills. This screen turns clinical directives into an achievable daily routine.

KEY FEATURES

- **Digital Checkbox:** Tap to toggle status with instantaneous timestamp.
- **Meal Composition:** Explicitly states low-salt / diabetic instructions.
- **Caregiver Verification:** Prevents missed nutrition or hydration gaps.

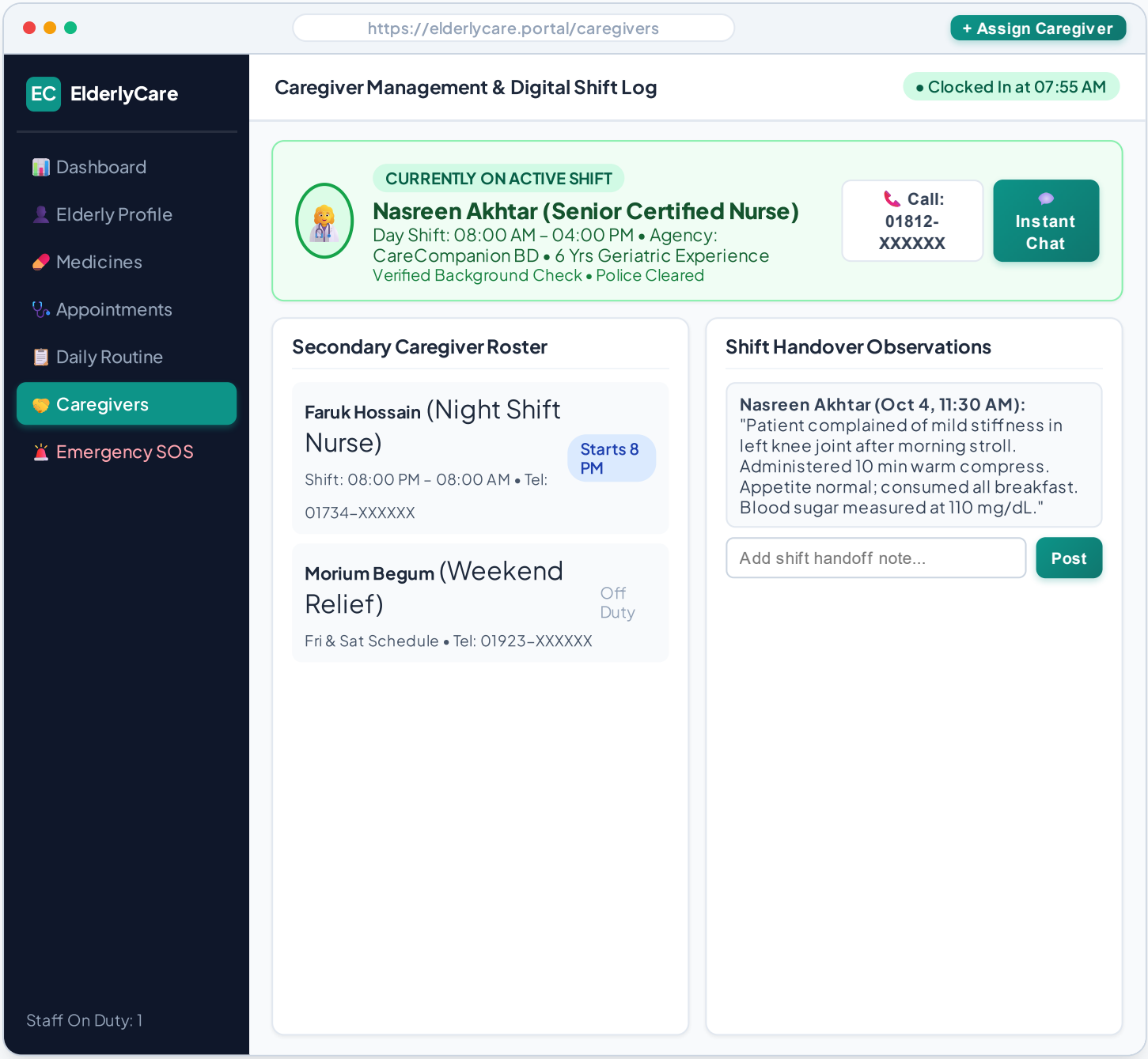
CATEGORIES

Nutrition | Vitals | Hydration | Mobility

10. Screen 8: Caregiver Information & Shift Handoff Log

Caregiver directory, verified credentials, shift clock-in, duty checklist, and observation exchange

Screen 8 of 12 | Caregiver Coordination



SCREEN PURPOSE

Facilitates smooth clinical handoffs between morning and night caregivers while keeping family members informed of daily behavioral changes.

KEY FEATURES

- **Direct Connectivity:** Family can phone or WhatsApp the caregiver with 1 tap.
- **Verified Staff Credentials:** Displays nurse training and police clearance.
- **Observations Log:** Real-time logging of physical discomfort or symptoms.

ACCOUNTABILITY

GPS Geofence | Clock In/Out | Handoff Notes

11. Screen 9: Instant Emergency SOS & Paramedic Brief

Rapid one-touch emergency dispatch, hospital ER direct routing, and immediate medical summary

Critical Safety Protocol | Screen 9 of 12

https://elderlycare.portal/emergency-sos

RAPID RESPONSE ACTIVE

SOS ElderlyCare

- Dashboard
- Elderly Profile
- Medicines
- Appointments
- Daily Routine
- Caregivers
- Emergency SOS

Hotline: 999 / 10616

IMMEDIATE EMERGENCY BROADCAST CONSOLE

TRIGGER BROADCAST SMS TO ALL FAMILY

National Ambulance

999 / 199

CALL AMBULANCE NOW

Square Hospital ER

10616 / 02-8144400

CALL ER DESK NOW

Dr. Hasan (Cardio)

01713-XXXXXX

CALL DOCTOR DIRECT

Family Emergency Contacts (Priority Order)

Tariq Rahman

Son • Priority 1

Lives 10 mins away • Tel: 01711-XXXXXX

Call 1st

Farzana Chowdhury

Daughter • Priority 2

Tel: 01819-XXXXXX • Alternative Contact

Call 2nd

Paramedic Critical Health Brief (On Arrival)

- **Patient:** Abdul Rahman Chowdhury (Age: 76 Years)
- **Blood Group:** B Positive (B+)
- **CRITICAL ALLERGY:** PENICILLIN (Causes Anaphylaxis)
- **Key Conditions:** Type 2 Diabetes, Hypertension, Coronary Stent (2021)
- **Home GPS Coordinates:** 23.7461° N, 90.3742° E (Dhanmondi Road 11)

CRITICAL LIFE-SAFETY PROTOCOL

In medical crises, every second counts. This screen provides massive single-touch speed dial triggers and broadcasts the senior's exact home location and vital history instantly.

EMERGENCY CAPABILITIES

- **One-Tap Speed Dial:** Direct links to National Ambulance and Hospital ER desk.
- **Paramedic Summary:** Prevents lethal drug reactions (Penicillin allergy clearly flagged).
- **Automated Family Ping:** Sends distress SMS with GPS pin to all relatives.

EMERGENCY TOKENS

#E11D48 Vivid Red | Speed Dial | GPS Lat/Long

SR

12. Screen 10: Senior Resident Accessible Dashboard

WCAG AAA dedicated senior portal featuring ≥14pt (18.67px) text, large ≥56px buttons, med reminders & 999 SOS

Screen 10 of 12 | Senior WCAG AAA UI



SENIOR ACCESSIBILITY

Built exclusively for aging eyes and motor-impaired hands using WCAG AAA high-contrast dark slate surfaces and oversized action controls.

TYPOGRAPHY & SAFETY

- **14pt (18.67px) Floor:** Zero microscopic text; all body reminders scale ≥14pt.
- **Large Touch Buttons:** Speed dial & SOS targets exceed 56px–76px height.
- **Accidental Tap Guard:** 5-second audible beep countdown before dispatching 999 SOS.

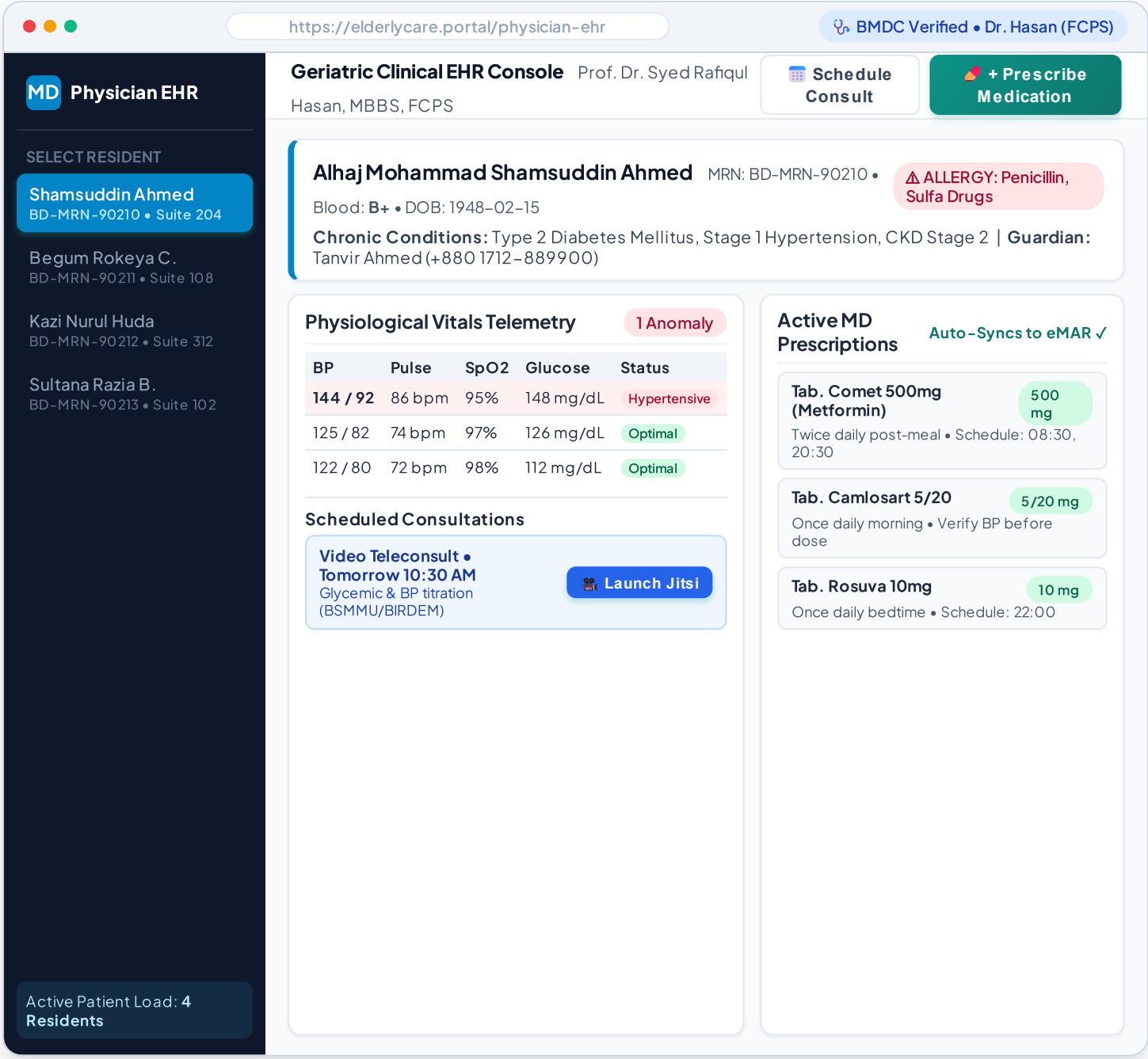
WCAG TOKENS

14pt / 18.67px Body | ≥56px Buttons | 7:1 Contrast

13. Screen 11: Physician Clinical EHR & Consultation Portal

Dedicated MD workspace for longitudinal EHR review, e-prescribing, allergy checks & video teleconsultations

Screen 11 of 12 | DOCTOR Role Portal



CLINICAL MD AUTHORITY

Separates physician ordering workflows from nurse execution. Doctors review longitudinal trends before issuing new prescriptions.

PHYSICIAN CAPABILITIES

- **Allergy Guardrail:** High-contrast drug allergy banner pinned atop EHR header.
- **Order-to-eMAR Sync:** Prescribing creates timed `PENDING` slots for duty nurses.
- **Integrated Telehealth:** One-click encrypted Jitsi Meet video consultation rooms.

CLINICAL TOKENS

#0284C7 Medical Blue | EHR Timeline | Jitsi Telehealth

14. Screen 12: Administrator Governance, RBAC & Audit Dashboard

Enterprise system monitoring KPIs, user lifecycle provisioning, RBAC permissions matrix & HIPAA audit logs

Screen 12 of 12 | ADMIN Role Portal

https://elderlycare.portal/admin-governance

● Telemetry Socket Online

AD ECMS Admin

- System KPIs
- User Accounts
- Role Permissions
- HIPAA Audit Logs
- SOS Dispatches

Director
Engr. Tariqul Islam

National Elderly Care Administration & Audit Intelligence

⬇ Export Audit Trail (JSON) | + Add New User

TOTAL SENIORS	VITALS TELEMETRY	MAR ADHERENCE	EMERGENCY ALERTS
4 Active	5 (1 Anomaly)	96.4%	0 Active

RBAC System User Management (5 Accounts)

User	Role	Status	Control
Engr. Tariqul Islam	ADMIN	ACTIVE	Current Session
Prof. Dr. S. R. Hasan	DOCTOR	ACTIVE	Terminate
Fatema Begum (RN)	CAREGIVER	ACTIVE	Terminate
Tanvir Ahmed	GUARDIAN	ACTIVE	Terminate
Shamsuddin Ahmed	SENIOR	ACTIVE	Terminate

Role Permissions & HIPAA Audit Trail

Module	Admin	Doctor	Nurse	Senior	Family
EHR / Prescribe	✓	✓	MAR Only	Self View	Monitor
Users / Audit Log	✓ Full	—	—	—	—

USER_LOGIN: dr.hasan@probin.org.bd • Authenticated via JWT

VITALS_RECORDED: Fatema Begum • Anomaly Flagged (144/92)

SOS_RESOLVED: Emergency #1 closed with clinical note

GOVERNANCE & SECURITY

Enforces strict RBAC least-privilege isolation, account lifecycle termination/reactivation, and immutable compliance auditing.

ADMIN CAPABILITIES

- **System Monitoring:** Real-time resident census, vitals anomalies & MAR rate.
- **Self-Lockout Guard:** Admin cannot terminate their own active session.
- **Audit Export:** One-click downloadable HIPAA JSON compliance trail.

GOVERNANCE TOKENS

JWT RBAC | Audit JSON | System KPIs